VINAY PHARMAMES

ENTERPRISE PHARMACEUTICAL MANUFACTURING EXECUTION SYSTEM

21 CFR PART 11 ACTIVE

Project Technical Summary, Architecture & System Walkthrough Guide

AUTHOR & PLANT LEAD
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TECHNOLOGY STACK
Java 17, Spring Boot 3, Maven

DATA PERSISTENCE
Embedded H2 File Engine

GITHUB REPOSITORY
vinay0716yadav-ops/pharmames

1. The Executive 30-Second Elevator Pitch

> *"Vinay PharmaMES is an enterprise-grade digital Pharmaceutical Manufacturing Execution System built with Java 17 and Spring Boot 3. It replaces traditional paper-based batch records with a fully validated, Good Manufacturing Practice (GMP) digital workflow. It tracks pharmaceutical lots through the entire 7-stage manufacturing lifecycle, enforces FDA 21 CFR Part 11 immutable audit logging, captures Critical Process Parameters (CPPs), and permanently preserves records on physical disk."*

2. Pharmaceutical Industry Problem & Digital Solution

CHALLENGE AREA	TRADITIONAL PAPER BATCH RECORDS (PBR)	VINAY PHARMAMES DIGITAL SOLUTION
Data Integrity	Physical binders prone to illegible handwriting, missed sign-offs, and accidental spills.	Guaranteed data consistency, mandatory validation fields, and permanent disk commits.
Process Deviations	Temperature or speed excursions often discovered days later during paper review.	Immediate flagging of deviations during unit operations with mandatory root-cause notes.
Regulatory Audits	FDA and GMP inspectors spend weeks manually cross-referencing paper logs.	Instant, tamper-evident 21 CFR Part 11 audit trails with one-click print-ready dossiers.
Batch Release Time	Weeks required for QA to review paper stacks before shipping commercial goods.	Analytical QC assay results and QA release dispositions executed digitally in minutes.

3. The 7-Stage Pharmaceutical Unit Operations Pipeline

Every batch flows through strict Good Manufacturing Practice (GMP) unit operations:

1. Dispensing
Raw API & Excipient weighing

2. Granulation
High-shear wet & fluid bed

3. Compression
Rotary tablet press / filling

4. Coating
Aqueous film finish

5. Packaging
Blister & serialization

6. QC Testing
Assay & dissolution lab

7. Released
QA Market certification

4. Deep Dive: The 5 Core Functional Modules

1. Production Dashboard /DASHBOARD

Real-time executive oversight across the plant floor.

- **Fleet KPIs:** Live counts for Total, Active In-Progress, Completed, Pending, and Quarantined batches.
- **Chart.js Analytics:** Interactive status doughnut chart and unit operations density bar chart.
- **Live Regulatory Feed:** Streaming feed of operator actions, status updates, and sign-offs.
- **Storage Verification:** Header badge confirming permanent disk DB connectivity.

2. Batch Management /BATCHES

End-to-end master formula and lot lifecycle tracking.

- **Order Initiation:** Generate lots specifying dosage forms (Tablets, Capsules, Liquids, Injectables), quantities, and lines.
- **Lifecycle Advancement:** Controlled stage transitions with supervisor sign-off.
- **Quarantine Control:** Instant ability to place lots on hold for investigation or resume production.
- **Export Dossier:** One-click print-ready batch dossier for inspectors.

3. Electronic Batch Records /EBR

Digital execution of master manufacturing operations.

- **Critical Parameters (CPPs):** Real-time comparison of Target vs. Actual measured temperatures, pressures, and RPM.
- **Dual Signatures:** Operator execution signature and supervisor verification sign-off.
- **Excursion Logging:** Steps flagged as PASS, DEVIATION, or FAIL with mandatory explanation.
- **Eliminates Paper:** Zero handwritten errors on the cleanroom production floor.

4. Quality Control & Release /QUALITY

Analytical lab testing & formal QA market release.

- **Analytical Assays:** Verifies Active Ingredient Assay % (95-105%), Dissolution Rate % (≥80%), and Moisture (≤3%).
- **Microbial Verification:** Confirms bioburden compliance with USP/EP compendial standards.
- **QA Release Disposition:** Authorizes commercial distribution (APPROVED) or rejects non-conforming product (REJECTED).

5. Regulatory Audit Trail (FDA 21 CFR Part 11 Enforced) /AUDIT

The regulatory backbone of pharmaceutical software ensuring legal traceability and non-repudiation.

- **Computer-Generated Timestamps:** Exact UTC timestamps recorded for every batch action.
- **User Identity & Role:** Logs the exact user (e.g., *Vinay Yadav - Formulation Lead*).
- **Reason for Change:** Every status transition, stage advance, or parameter sign-off requires an explicit operational reason.
- **Immutable Append-Only Log:** Audit records cannot be altered, overwritten, or erased by any user.

5. Permanent Disk Persistence Architecture

ATTRIBUTE	CONFIGURATION SPECIFICATION	TECHNICAL BENEFIT
Database Engine	H2 Embedded Persistent Engine with AUTO_SERVER=TRUE	Zero external database setup required; lightweight and blazingly fast.
File Storage Path	`./data/pharmames_db.mv.db`	Written directly to disk; data persists permanently across application reboots.
Transaction Safety	Spring Data JPA with Hibernate DDL `update` mode	Ensures transactional ACID guarantees on every batch record and audit row.
Inspection Console	Web Console accessible at `http://localhost:8080/h2-console`	Allows authorized database administrators to inspect tables and raw SQL directly.

6. Script for a 3-Minute Live System Demonstration

STEP	WHERE TO CLICK	WHAT TO SAY DURING THE PRESENTATION
Step 1	`/dashboard`	“Notice our real-time executive dashboard. The KPI cards show total, active, and completed lots. The Chart.js doughnut chart visualizes batch distribution, and the storage banner confirms permanent disk persistence.”
Step 2	`/batches` → `BAT-2026-001`	“This is a completed batch of Paracetamol 500mg Tablets. The 7-stage stepper shows 100% completion. Below, you see the executed EBR steps with measured parameters, QC assay results, and the batch audit trail.”
Step 3	`BAT-2026-002` → **Advance**	“Batch 002 is currently in Granulation. With one click on 'Advance to Next Stage', it moves into Compression, updates the visual stepper, and automatically generates a 21 CFR Part 11 audit entry.”
Step 4	`/ebr/new`	“Here on the shop floor, operators record critical process parameters like temperature and RPM. We enter the measured values, electronically sign as Vinay Yadav, and commit the step.”
Step 5	`/quality` → **Review**	“Before any medication reaches the market, QC inputs analytical lab assays. Once the QA Director signs with 'APPROVED', the batch is formally certified for commercial distribution.”
Step 6	`/audit`	“Every single action we just performed has been logged to this immutable, tamper-evident audit trail with timestamp, user identity, and reason, ready for an FDA inspection.”

7. Technology Stack & DevOps Reference

COMPONENT	TECHNOLOGY	IMPLEMENTATION DETAILS
Language & Runtime	Java 17 LTS	Targeted bytecode compilation, strong type safety, modern records and pattern matching.
Application Framework	Spring Boot 3.2.5	Spring MVC, Spring Data JPA, Hibernate ORM, and Jakarta Bean Validation.
Frontend Architecture	Thymeleaf + Chart.js	Fast server-side HTML rendering, modern CSS3 clinical theme, and interactive charts.
Containerization	Docker & Docker Compose	Multi-stage Dockerfile packaging into a secure, lightweight Alpine JRE non-root container.
CI/CD Automation	GitHub Actions	Automated workflow compiling code, executing automated test suites, and building container images.

8. Quick Command Cheat-Sheet

```
# Run application from project directory
cd /Users/apple/vinay/pharmames && mvn spring-boot:run

# Run automated unit & integration tests
mvn clean test

# Launch directly in Visual Studio Code (Press F5 to debug)
open -a "Visual Studio Code" /Users/apple/vinay/pharmames

# Build & run with Docker container
docker build -t pharmames-dashboard:latest .
docker run -d -p 8080:8080 -v $(pwd)/data:/app/data --name pharmames-app pharmames-dashboard:latest
```